

REPORT ANALYSIS OF SANGER ANALYSIS, BLAST, ALIGNMENT, AND PHYLOGENETICS

Sample Information

Sample Name : 92409060308/ By Ny Chodijah, 92409060292/ By Ny evi Ratnawati, 92411080331/ By Ny Pemilusian, 925011808806/ By Ny Hera Haerani, 92501200395/ By Ny Reni Puspitasari, 92503180167/ By Ny Jawati, 92505620648/ By Ny Ririn Amelia, 925061000356/ By Ny Dede Yunengsih, 92507150459/ By Ny Nurul P, 92507280163/ By Ny Natalia P, 92508040479 / by By Ny Ina Kushandayani, 92509160007/ By Ny Anggie Sukma, and 92508130363/ By Ny ArumSari.

Sample Number : 511.S.3 - 511.S.16

Marketing Number : SIG.MARK.S.XI.2025.000119

Test Method : 18-24-13/MU

Date Received : 07/11/2025

Date Tested : 14/11/2025

Date Reported : 09/12/2025

A. Methods

Genomic DNA was extracted from blood samples using the Quick-DNA Miniprep Plus Kit, following the manufacturer's protocol. PCR amplification was carried out using the primer pair 4_144_F and 4_833_R, targeting a fragment of approximately ~700 bp. The PCR products, which showed a single clear band, were subjected to Sanger sequencing in both forward and reverse directions for each sample. The resulting forward and reverse sequences were quality-checked and assembled into consensus sequences using DNA Baser Assembler. The consensus sequences were then analyzed using BLASTn (megablast option) against the NCBI nr database (<https://blast.ncbi.nlm.nih.gov>). Multiple sequence alignment of the consensus sequences and selected reference sequences was performed using MEGA software with the ClustalW algorithm. Phylogenetic analysis was subsequently conducted based on the aligned sequences using the Neighbor-Joining (NJ) method. The Tamura-Nei substitution model was applied, and bootstrap analysis was performed to evaluate the reliability of the resulting tree.

B. Results

1. DNA Extraction

Table 1. DNA extraction result

No.	Sample Number	Sample Name	Concentration (ng/μL)
1	511.S.3	By Ny Chodijah	5.02
2	511.S.4	By Ny evi Ratnawati	4.51
3	511.S.6	By Ny Pemilusian	5.33
4	511.S.7	By Ny Hera Haerani	6.77

5	511.S.8	By Ny Reni Puspitasari	5.28
6	511.S.9	By Ny Jawati	12.87
7	511.S.10	By Ny Ririn Amelia	7.57
8	511.S.11	By Ny Dede Yunengsih	57.97
9	511.S.12	By Ny Nurul P	13.81
10	511.S.13	By Ny Natalia P	10.61
11	511.S.14	By Ny Ina Kushandayani	15.98
12	511.S.15	By Ny Anggie Sukma	11.54
13	511.S.16	By Ny Arum Sari	7.65

2. PCR

Table 2. Primer information

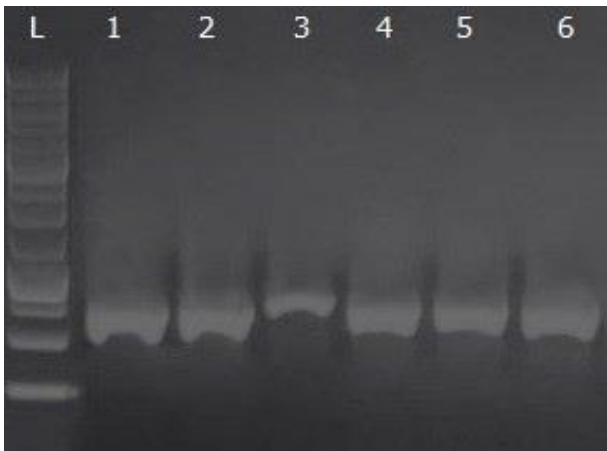
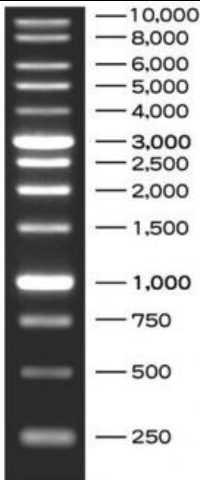
No.	Primer	Sequences (5'→3')	Expected Product Size
1	4_144_F	TTC AGC CCC ATC TTA GCA GC	~ 700
2	4_833_R	GTC CCC AGC CAC TTC TAA CC	

Table 3. PCR Profile

Step	Temperature (°C)	Time	Cycle(s)
Initial Denaturation	98°C	1 minutes	1 cycle
Denaturation	98°C	10 seconds	35 cycles
Annealing	62.3°C	5 seconds	
Extension	68°C	18 seconds	
Final Extension	68°C	5 minutes	1 cycle
Hold	4°C	∞	-

3. Gel Electrophoresis Image of PCR Amplification

Table 4. Electrophoresis result

Figure	Description	Ladder Reference
	L : 1 Kb DNA ladder 1 : 511.S.3 2 : 511.S.4 3 : 511.S.6 4 : 511.S.7 5 : 511.S.8 6 : 511.S.9 7 : 511.S.10 8 : 511.S.11 9 : 511.S.12 10: 511.S.13 11: 511.S.14 12: 511.S.15 13: 511.S.16	
		

4. Sanger Sequencing

Table 5. Sequencing result

No.	Sample Code	Sample Name	Primer Name	Sample Type	Target size (bp)	Sample Concentration	Sequencing result
1	511.S.3	By Ny Chodijah	4_144_F	PCR product	700	Normal., Sequencing	Success
			4_833_R			Normal., Sequencing	Success
2	511.S.4	By Ny evi Ratnawati	4_144_F	PCR product	700	Normal., Sequencing	Success
			4_833_R			Normal., Sequencing	Success
3	511.S.6	By Ny Pemilusan	4_144_F	PCR product	700	Normal., Sequencing	Success
			4_833_R			Normal., Sequencing	Success
4	511.S.7		4_144_F	PCR product	700	Normal., Sequencing	Success

		By Ny Hera Haerani	4_833_R			Normal., Sequencing	Success
5	511.S.8	By Ny Reni Puspitasari	4_144_F	PCR product	700	Normal., Sequencing	Success
			4_833_R			Normal., Sequencing	Success
6	511.S.9	By Ny Jawati	4_144_F	PCR product	700	Normal., Sequencing	Success
			4_833_R			Normal., Sequencing	Success
7	511.S.10	By Ny Ririn Amelia	4_144_F	PCR product	700	Normal., Sequencing	Success
			4_833_R			Normal., Sequencing	Success
8	511.S.11	By Ny Dede Yunengsih	4_144_F	PCR product	700	Normal., Sequencing	Success
			4_833_R			Normal., Sequencing	Success
9	511.S.12	By Ny Nurul P	4_144_F	PCR product	700	Normal., Sequencing	Success
			4_833_R			Normal., Sequencing	Success
10	511.S.13	By Ny Natalia P	4_144_F	PCR product	700	Normal., Sequencing	Success
			4_833_R			Normal., Sequencing	Success
11	511.S.14	By Ny Ina Kushandayani	4_144_F	PCR product	700	Normal., Sequencing	Success
			4_833_R			Normal., Sequencing	Success
12	511.S.15	By Ny Anggie Sukma	4_144_F	PCR product	700	Normal., Sequencing	Success
			4_833_R			Normal., Sequencing	Success
13	511.S.16	By Ny ArumSari	4_144_F	PCR product	700	Normal., Sequencing	Success
			4_833_R			Normal., Sequencing	Success

5. Data Analysis BLAST Result by NCBI

Table 6. Top 10 Hits Analysis of By Ny Chodijah

Description	Scientific Name	Max Score	Query Cover	E value	Per. Ident (%)	Accession
<i>Homo sapiens</i> isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1027	99%	0	99.64	DQ173610.1
<i>Homo sapiens</i> isolate YCC22 glucose-6-phosphate	<i>Homo sapiens</i>	1027	99%	0	99.64	AY158101.1

dehydrogenase (G6PD) gene, partial cds						
<i>Homo sapiens</i> isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1027	99%	0	99.64	DQ173614.1
<i>Homo sapiens</i> isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1027	99%	0	99.64	DQ173592.1
<i>Homo sapiens</i> isolate SHO_14_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1027	99%	0	99.64	DQ173596.1
<i>Homo sapiens</i> isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1027	99%	0	99.64	DQ173608.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1027	99%	0	99.64	ON262108.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD)	<i>Homo sapiens</i>	1027	99%	0	99.64	NG_009015.2
<i>Homo sapiens</i> isolate YCC10 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1027	99%	0	99.64	AY158129.1
<i>Homo sapiens</i> isolate YCC67 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1027	99%	0	99.64	AY158134.1

Table 7. Top 10 Hits Analysis of By Ny evi Ratnawati

Description	Scientific Name	Max Score	Query Cover	E value	Per. Indent (%)	Accession
<i>Homo sapiens</i> isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1033	100%	0	100	DQ173610.1
<i>Homo sapiens</i> isolate YCC22 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1033	100%	0	100	AY158101.1
<i>Homo sapiens</i> isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1033	100%	0	100	DQ173614.1

<i>Homo sapiens</i> isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1033	100%	0	100	DQ173592.1
<i>Homo sapiens</i> isolate SHO_14_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1033	100%	0	100	DQ173596.1
<i>Homo sapiens</i> isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1033	100%	0	100	DQ173608.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1033	100%	0	100	ON262108.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD), RefSeqGene on chromosome X	<i>Homo sapiens</i>	1033	100%	0	100	NG_009015.2
<i>Homo sapiens</i> isolate YCC10 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1033	100%	0	100	AY158129.1
<i>Homo sapiens</i> isolate YCC67 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1033	100%	0	100	AY158134.1

Table 8. Top 10 Hits Analysis of By Ny Pemilusan

Description	Scientific Name	Max Score	Query Cover	E value	Per. Ident (%)	Accession
<i>Homo sapiens</i> isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	100	DQ173610.1
<i>Homo sapiens</i> isolate YCC22 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1037	100%	0	100	AY158101.1
<i>Homo sapiens</i> isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	100	DQ173614.1
<i>Homo sapiens</i> isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	100	DQ173592.1
<i>Homo sapiens</i> isolate SHO_14_B biglycan	<i>Homo sapiens</i>	1037	100%	0	100	DQ173596.1

(BGN) gene, partial sequence						
<i>Homo sapiens</i> isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	100	DQ173608.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1037	100%	0	100	ON262108.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD), RefSeqGene on chromosome X	<i>Homo sapiens</i>	1037	100%	0	100	NG_009015.2
<i>Homo sapiens</i> isolate YCC10 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1037	100%	0	100	AY158129.1
<i>Homo sapiens</i> isolate YCC67 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1037	100%	0	100	AY158134.1

Table 9. Top 10 Hits Analysis of By Ny Hera Haerani

Description	Scientific Name	Max Score	Query Cover	E value	Per. Indent (%)	Accession
<i>Homo sapiens</i> isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1031	100%	0	100	DQ173610.1
<i>Homo sapiens</i> isolate YCC22 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1031	100%	0	100	AY158101.1
<i>Homo sapiens</i> isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1031	100%	0	100	DQ173614.1
<i>Homo sapiens</i> isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1031	100%	0	100	DQ173592.1
<i>Homo sapiens</i> isolate SHO_14_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1031	100%	0	100	DQ173596.1
<i>Homo sapiens</i> isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1031	100%	0	100	DQ173608.1

<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1031	100%	0	100	ON262108.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD), RefSeqGene on chromosome X	<i>Homo sapiens</i>	1031	100%	0	100	NG_009015.2
<i>Homo sapiens</i> isolate YCC10 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1031	100%	0	100	AY158129.1
<i>Homo sapiens</i> isolate YCC67 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1031	100%	0	100	AY158134.1

Table 10. Top 10 Hits Analysis of By Ny Reni Puspitasari

Description	Scientific Name	Max Score	Query Cover	E value	Per. Indent (%)	Accession
<i>Homo sapiens</i> isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	100	DQ173610.1
<i>Homo sapiens</i> isolate YCC22 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1037	100%	0	100	AY158101.1
<i>Homo sapiens</i> isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	100	DQ173614.1
<i>Homo sapiens</i> isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	100	DQ173592.1
<i>Homo sapiens</i> isolate SHO_14_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	100	DQ173596.1
<i>Homo sapiens</i> isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	100	DQ173608.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1037	100%	0	100	ON262108.1

<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD), RefSeqGene on chromosome X	<i>Homo sapiens</i>	1037	100%	0	100	NG_009015.2
<i>Homo sapiens</i> isolate YCC10 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1037	100%	0	100	AY158129.1
<i>Homo sapiens</i> isolate YCC67 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1037	100%	0	100	AY158134.1

Table 11. Top 10 Hits Analysis of By Ny Jawati

Description	Scientific Name	Max Score	Query Cover	E value	Per. Indent (%)	Accession
<i>Homo sapiens</i> isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1011	100%	0	100	DQ173610.1
<i>Homo sapiens</i> isolate YCC22 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1011	99%	0	100	AY158101.1
<i>Homo sapiens</i> isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1011	100%	0	100	DQ173614.1
<i>Homo sapiens</i> isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1011	100%	0	100	DQ173592.1
<i>Homo sapiens</i> isolate SHO_14_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1011	100%	0	100	DQ173596.1
<i>Homo sapiens</i> isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1011	100%	0	100	DQ173608.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1011	99%	0	100	ON262108.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD),	<i>Homo sapiens</i>	1011	99%	0	100	NG_009015.2

RefSeqGene on chromosome X						
<i>Homo sapiens</i> isolate YCC10 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1011	99%	0	100	AY158129.1
<i>Homo sapiens</i> isolate YCC67 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1011	99%	0	100	AY158134.1

Table 12. Top 10 Hits Analysis of By Ny Ririn Amelia

Description	Scientific Name	Max Score	Query Cover	E value	Per. Indent (%)	Accession
<i>Homo sapiens</i> isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1011	100%	0	100	DQ173610.1
<i>Homo sapiens</i> isolate YCC22 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1011	100%	0	100	AY158101.1
<i>Homo sapiens</i> isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1011	100%	0	100	DQ173614.1
<i>Homo sapiens</i> isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1011	100%	0	100	DQ173592.1
<i>Homo sapiens</i> isolate SHO_14_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1011	100%	0	100	DQ173596.1
<i>Homo sapiens</i> isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1011	100%	0	100	DQ173608.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1011	100%	0	100	ON262108.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD), RefSeqGene on chromosome X	<i>Homo sapiens</i>	1011	100%	0	100	NG_009015.2
<i>Homo sapiens</i> isolate YCC10 glucose-6-	<i>Homo sapiens</i>	1011	100%	0	100	AY158129.1

phosphate dehydrogenase (G6PD) gene, partial cds						
<i>Homo sapiens</i> isolate YCC67 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1011	100%	0	100	AY158134.1

Table 13. Top 10 Hits Analysis of By Ny Dede Yunengsih

Description	Scientific Name	Max Score	Query Cover	E value	Per. Indent (%)	Accession
<i>Homo sapiens</i> isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	99.82	DQ173610.1
<i>Homo sapiens</i> isolate YCC22 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1037	100%	0	99.82	AY158101.1
<i>Homo sapiens</i> isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	99.82	DQ173614.1
<i>Homo sapiens</i> isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	99.82	DQ173592.1
<i>Homo sapiens</i> isolate SHO_14_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	99.82	DQ173596.1
<i>Homo sapiens</i> isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1037	100%	0	99.82	DQ173608.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1037	100%	0	99.82	ON262108.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD), RefSeqGene on chromosome X	<i>Homo sapiens</i>	1037	100%	0	99.82	NG_009015.2
<i>Homo sapiens</i> isolate YCC10 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1037	100%	0	99.82	AY158129.1

<i>Homo sapiens</i> isolate YCC67 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1037	100%	0	99.82	AY158134.1
--	---------------------	------	------	---	-------	------------

Table 14. Top 10 Hits Analysis of By Ny Nurul P

Description	Scientific Name	Max Score	Query Cover	E value	Per. Indent (%)	Accession
<i>Homo sapiens</i> isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1055	100%	0	99.83	DQ173610.1
<i>Homo sapiens</i> isolate YCC22 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1055	100%	0	99.83	AY158101.1
<i>Homo sapiens</i> isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1055	100%	0	99.83	DQ173614.1
<i>Homo sapiens</i> isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1055	100%	0	99.83	DQ173592.1
<i>Homo sapiens</i> isolate SHO_14_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1055	100%	0	99.83	DQ173596.1
<i>Homo sapiens</i> isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1055	100%	0	99.83	DQ173608.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1055	100%	0	99.83	ON262108.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD), RefSeqGene on chromosome X	<i>Homo sapiens</i>	1055	100%	0	99.83	NG_009015.2
<i>Homo sapiens</i> isolate YCC10 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1055	100%	0	99.83	AY158129.1
<i>Homo sapiens</i> isolate YCC67 glucose-6-phosphate	<i>Homo sapiens</i>	1055	100%	0	99.83	AY158134.1

dehydrogenase (G6PD) gene, partial cds						
---	--	--	--	--	--	--

Table 15. Top 10 Hits Analysis of By Ny Natalia P

Description	Scientific Name	Max Score	Query Cover	E value	Per. Indent (%)	Accession
<i>Homo sapiens</i> isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1044	99%	0	99.65	DQ173610.1
<i>Homo sapiens</i> isolate YCC22 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1044	99%	0	99.65	AY158101.1
<i>Homo sapiens</i> isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1044	99%	0	99.65	DQ173614.1
<i>Homo sapiens</i> isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1044	99%	0	99.65	DQ173592.1
<i>Homo sapiens</i> isolate SHO_14_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1044	99%	0	99.65	DQ173596.1
<i>Homo sapiens</i> isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1044	99%	0	99.65	DQ173608.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1044	99%	0	99.65	ON262108.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD), RefSeqGene on chromosome X	<i>Homo sapiens</i>	1044	99%	0	99.65	NG_009015.2
<i>Homo sapiens</i> isolate YCC10 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1044	99%	0	99.65	AY158129.1
<i>Homo sapiens</i> isolate YCC67 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1044	99%	0	99.65	AY158134.1

Table 16. Top 10 Hits Analysis of By Ny Ina Kushandayani

Description	Scientific Name	Max Score	Query Cover	E value	Per. Indent (%)	Accession
<i>Homo sapiens</i> isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1044	99%	0	99.65	DQ173610.1
<i>Homo sapiens</i> isolate YCC22 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1044	99%	0	99.65	AY158101.1
<i>Homo sapiens</i> isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1044	99%	0	99.65	DQ173614.1
<i>Homo sapiens</i> isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1044	99%	0	99.65	DQ173592.1
<i>Homo sapiens</i> isolate SHO_14_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1044	99%	0	99.65	DQ173596.1
<i>Homo sapiens</i> isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1044	99%	0	99.65	DQ173608.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1044	99%	0	99.65	ON262108.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD), RefSeqGene on chromosome X	<i>Homo sapiens</i>	1044	99%	0	99.65	NG_009015.2
<i>Homo sapiens</i> isolate YCC10 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1044	99%	0	99.65	AY158129.1
<i>Homo sapiens</i> isolate YCC67 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1044	99%	0	99.65	AY158134.1

Table 17. Top 10 Hits Analysis of By Ny Anggie Sukma

Description	Scientific Name	Max Score	Query Cover	E value	Per. Indent (%)	Accession
-------------	-----------------	-----------	-------------	---------	-----------------	-----------

<i>Homo sapiens</i> isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1031	100%	0	100	DQ173610.1
<i>Homo sapiens</i> isolate YCC22 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1031	100%	0	100	AY158101.1
<i>Homo sapiens</i> isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1031	100%	0	100	DQ173614.1
<i>Homo sapiens</i> isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1031	100%	0	100	DQ173592.1
<i>Homo sapiens</i> isolate SHO_14_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1031	100%	0	100	DQ173596.1
<i>Homo sapiens</i> isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1031	100%	0	100	DQ173608.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1031	100%	0	100	ON262108.1
<i>Homo sapiens</i> glucose-6-phosphate dehydrogenase (G6PD), RefSeqGene on chromosome X	<i>Homo sapiens</i>	1031	100%	0	100	NG_009015.2
<i>Homo sapiens</i> isolate YCC10 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1031	100%	0	100	AY158129.1
<i>Homo sapiens</i> isolate YCC67 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1031	100%	0	100	AY158134.1

Table 18. Top 10 Hits Analysis of By Ny ArumSari

Description	Scientific Name	Max Score	Query Cover	E value	Per. Indent (%)	Accession
Homo sapiens isolate MKA_21_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1221	100%	0	100	DQ173610.1
Homo sapiens isolate YCC22 glucose-6-	<i>Homo sapiens</i>	1221	100%	0	100	AY158101.1

phosphate dehydrogenase (G6PD) gene, partial cds						
Homo sapiens isolate JK_785_A+ biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1221	100%	0	100	DQ173614.1
Homo sapiens isolate KOH_188_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1221	100%	0	100	DQ173592.1
Homo sapiens isolate SHO_14_B biglycan (BGN) gene, partial sequence;	<i>Homo sapiens</i>	1221	100%	0	100	DQ173596.1
Homo sapiens isolate IVC_20_B biglycan (BGN) gene, partial sequence	<i>Homo sapiens</i>	1221	100%	0	100	DQ173608.1
Homo sapiens glucose-6-phosphate dehydrogenase (G6PD) gene, complete cds	<i>Homo sapiens</i>	1221	100%	0	100	ON262108.1
Homo sapiens glucose-6-phosphate dehydrogenase (G6PD), RefSeqGene on chromosome X	<i>Homo sapiens</i>	1221	100%	0	100	NG_009015.2
Homo sapiens isolate YCC10 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1221	100%	0	100	AY158129.1
Homo sapiens isolate YCC67 glucose-6-phosphate dehydrogenase (G6PD) gene, partial cds	<i>Homo sapiens</i>	1221	100%	0	100	AY158134.1

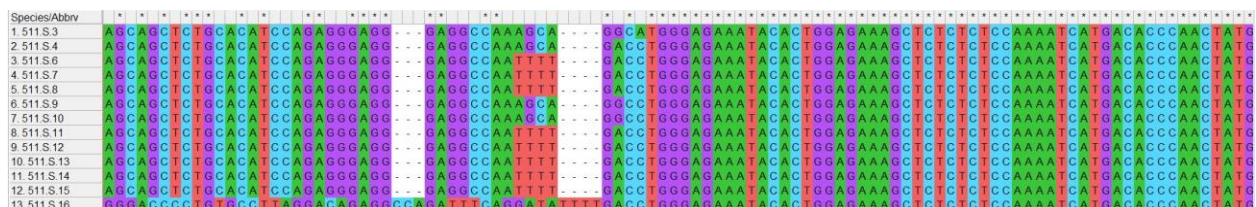


Figure 1. Multiple sequence alignment of the samples.

6. Phylogenetic Tree

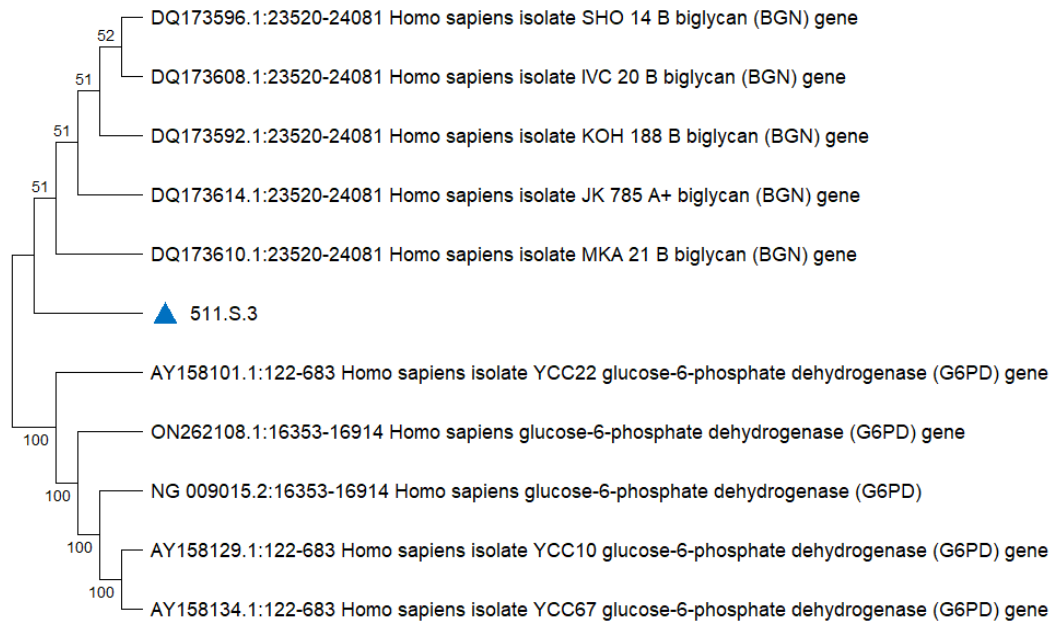


Figure 2. Phylogenetic Tree of By Ny Chodijah Samples using NJ Method.



Figure 3. Phylogenetic Tree of By Ny evi Ratnawati Samples using NJ Method.



Figure 4. Phylogenetic Tree of By Ny Pemilusan Samples using NJ Method.

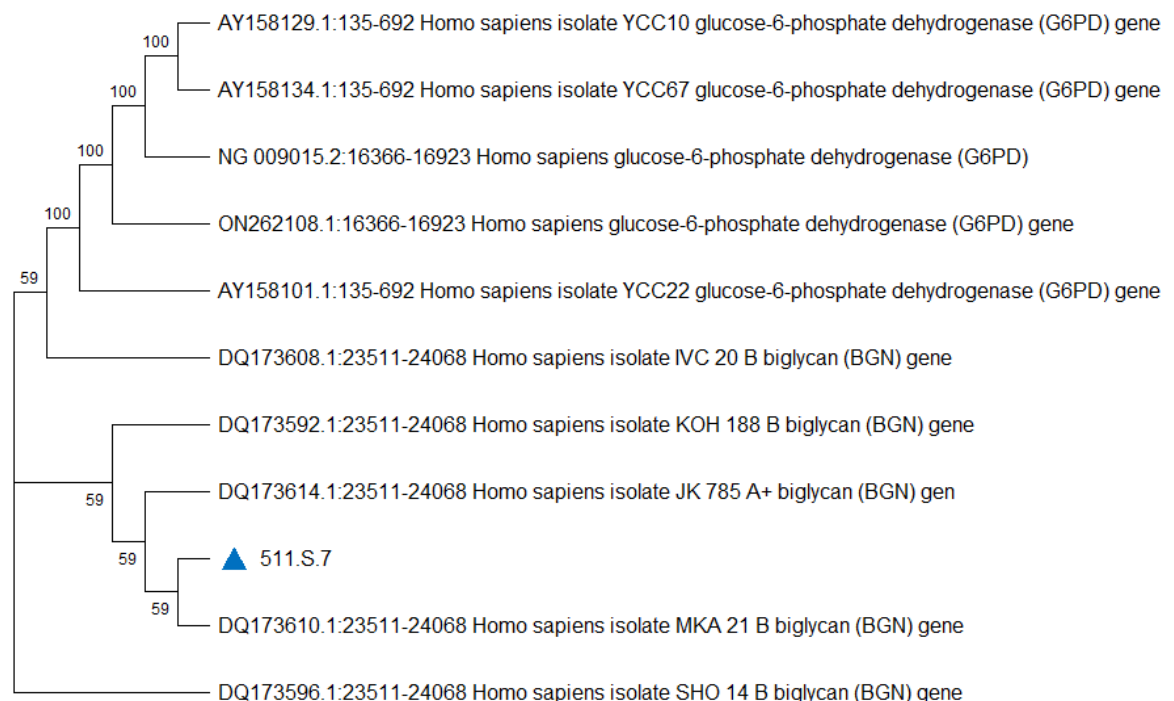


Figure 5. Phylogenetic Tree of By Ny Hera Haerani Samples using NJ Method.



Figure 6. Phylogenetic Tree of By Ny Reni Puspitasari Samples using NJ Method.



Figure 7. Phylogenetic Tree of By Ny Jawati Samples using NJ Method.



Figure 8. Phylogenetic Tree of By Ny Ririn Amelia Samples using NJ Method.

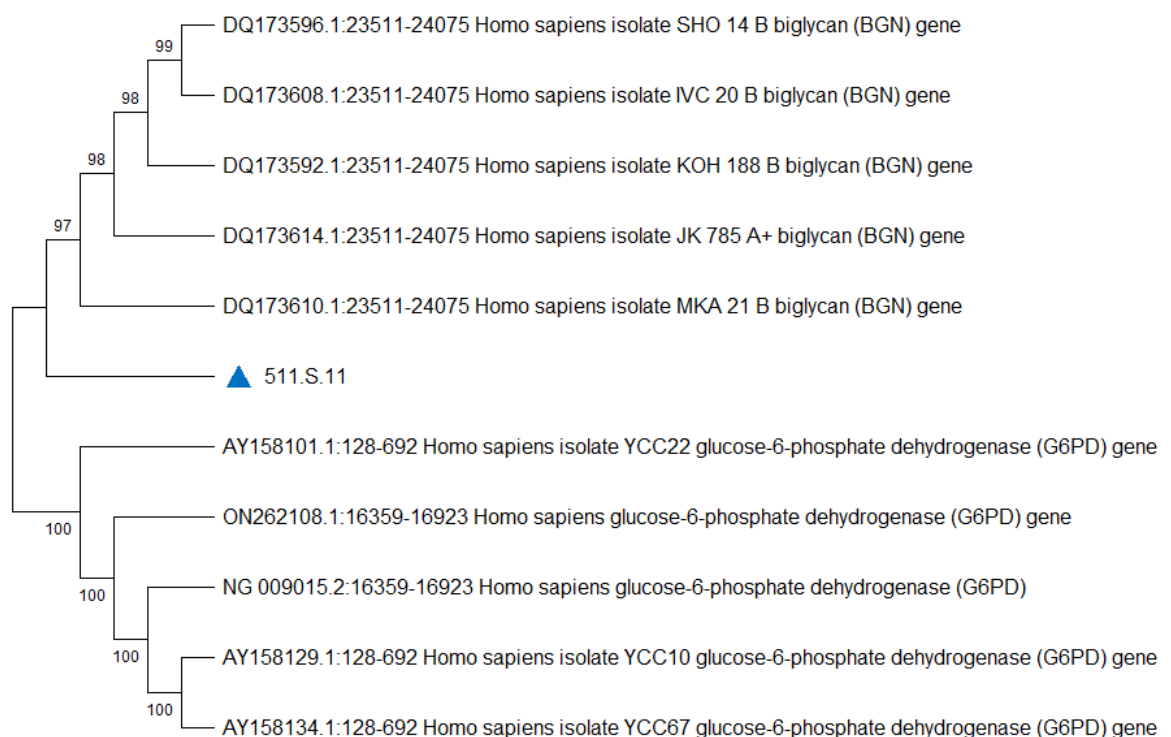


Figure 9. Phylogenetic Tree of By Ny Dede Yunengsih Samples using NJ Method.

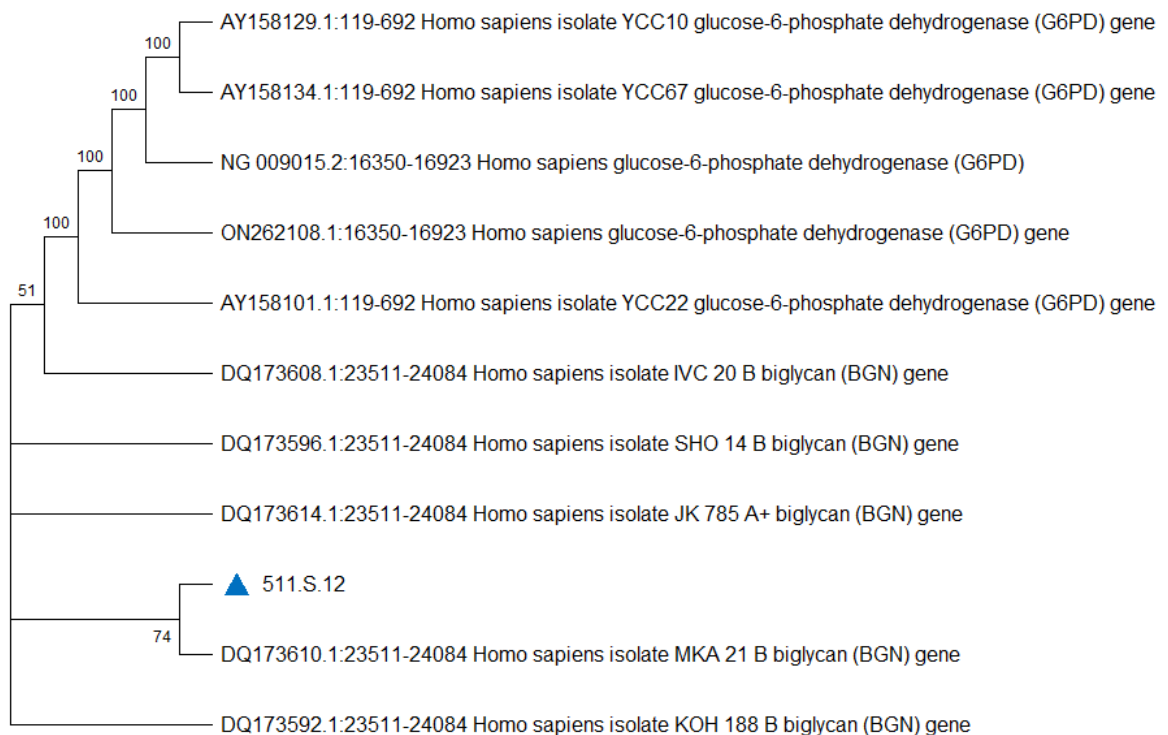


Figure 10. Phylogenetic Tree of By Ny Nurul P Samples using NJ Method.



Figure 11. Phylogenetic Tree of By Ny Natalia P Samples using NJ Method.



Figure 12. Phylogenetic Tree of By Ny Ina Kushandayani Samples using NJ Method.



Figure 13. Phylogenetic Tree of By Ny Angie Sukma Samples using NJ Method.



Figure 14. Phylogenetic Tree of By Ny Arum Sari Samples using NJ Method.

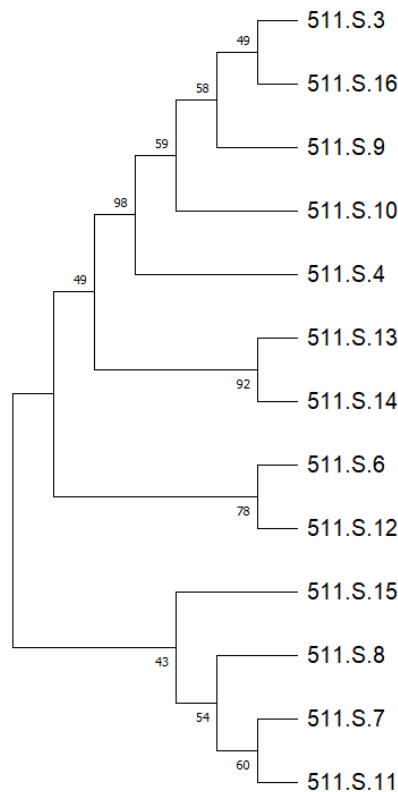


Figure 15. Phylogenetic tree of all analyzed samples.

Table 19. Raw data file

File	Description
.ab1 file	Chromatogram with quality value (QV). This file can be opened by Sequencing Analysis Software v5 (ABI), Sequence Scanner Software (ABI), Chromas and other versions suitable to the .ab1 file, however, the QV cannot be shown.
.seq file	Text file contains DNA sequence only. This file can be opened by common word processing software such as Microsoft Word & Notepad.
Contig	Assembled sequences in fasta format file.
.mas	Alignment sequence in fasta format file
.mtsx	Phylogenetics tree generated using Mega. The file can be opened using Mega software.
.png	Phylogenetics tree figure.

Reported by



Nurul Afia Abd Majid

Lab. Staff